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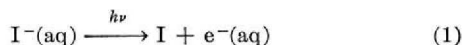
The Photochemical Oxidation of Aqueous Iodide Solutions

*An experiment demonstrating the competitive
reactions of the hydrated electron*

This article describes a teaching laboratory experiment in which hydrated electrons are formed by the photochemical method referred to in the preceding article (1). The experiment may be performed in about six hours in an undarkened room and needs very little specialized equipment. De-aeration of solutions and analysis of gaseous products, which are frequently necessary in this field, are not required. The results illustrate some of the reactions of the hydrated electron and permit the estimation of the *relative* rates of its reactions with two solutes. This experiment was designed for students in the final year of the Honours Chemistry Course at the University of Leeds.

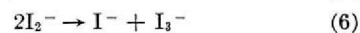
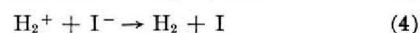
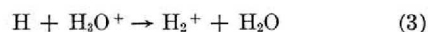
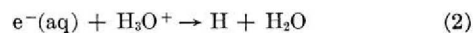
Theory

When aqueous solutions containing iodide ions are exposed to ultraviolet radiation of wavelength less than 2700 Å, hydrated electrons are formed by processes which may be summarized by eqn. (1).



The primary quantum yield of hydrated electrons, $\phi \text{e}^-(\text{aq})$,¹ is a function of temperature and wavelength and has been found to be 0.24 at 25°C for light of wavelength 2537 Å (mercury resonance radiation) (2).

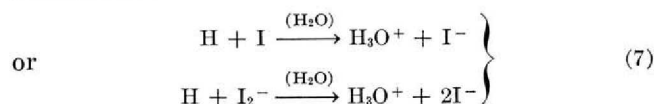
Among solutes which react rapidly with the hydrated electron are N_2O , H_3O^+ , and O_2 . In a deoxygenated iodide solution at natural pH, and in the absence of any other reactants, no net reaction is observed indicating that the primary products undergo recombination. As the solution is made more acidic a series of reactions occur which lead to the formation of I_3^- and H_2 .



¹ ϕ_A denotes the quantum yield of reactive intermediate, A, formed in the primary process. $\phi(X)$ denotes the measured quantum yield of a final product X.

At $pH < 3$ all of the primary electrons react with H_3O^+ to form hydrogen atoms and of these a fraction, f_1 , oxidize I^- via reactions (3) and (4).

The remainder undergo a back reaction with iodine atoms or I_2^- .



Under these conditions [remembering that an iodine atom is also formed in reaction (1)] the total quantum yield of I_3^- will be given by (8).

$$\phi(I_3^-) = \phi(H_2) = f_1 \cdot \phi_{e-(aq)} \quad (8)$$

In solutions of constant iodide concentration f_1 is a function of $[H_3O^+]$ and approaches unity at high acid concentrations. Thus in the absence of any competing solutes the iodine yield increases from almost zero in neutral solution to a value approaching that of the primary quantum yield, $\phi_{e-(aq)}$ in a strongly acid solution (see Fig. 1). At high conversions another back-reaction may become significant.

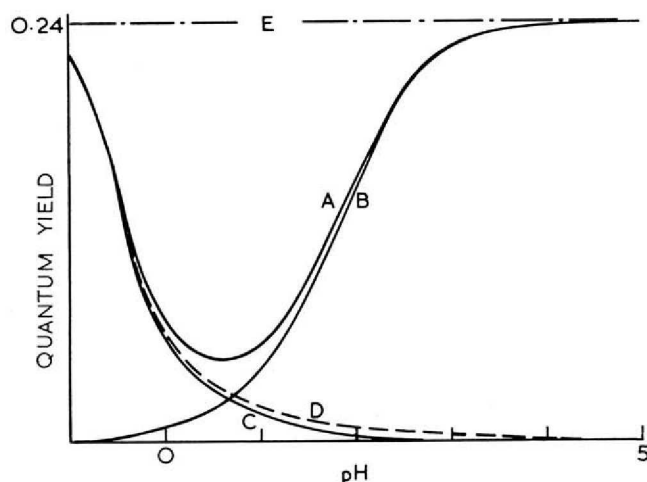
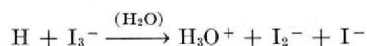
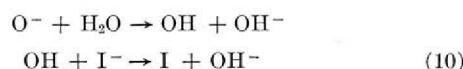


Figure 1. Quantum yields for the photolysis of aqueous solutions of NaI at 25°C with light of wavelength 2537 Å. Full lines refer to solutions saturated with N_2O : A, $\phi'(I_3^-)$; B, $\phi'(N_2)$; C, $\phi'(H_2)$. The broken line refers to deoxygenated solutions: D, $\phi(I_3^-) = \phi(H_2)$. Line E represents $\phi_{e-(aq)}$.

In a solution which contains N_2O a second series of reactions may occur which also leads to the formation of iodine.



(at $pH < 11$)



Thus in a solution of natural pH (in which no net reaction is observed in the absence of a solute undergoing an irreversible reaction with the photochemically generated intermediates) saturation with 1 atm N_2O causes all the primary electrons to undergo reaction (9) by which N_2O is reduced to N_2 and consequently iodine is obtained as the oxidized product. The stoichiometry is given by

$$\phi'(I_3^-) = \phi'(N_2) = \phi_{e-(aq)} \quad (11)$$

where ϕ' denotes the quantum yield of products from solutions saturated with N_2O .

As the acid concentration is increased there will be competition between N_2O and H_3O^+ for the electrons produced in reaction (1). The usual steady state treatment gives the fraction, f_N , of electrons which react with N_2O :

$$f_N = \frac{k_9[N_2O]}{k_9[N_2O] + k_2[H_3O^+]} \quad (12)$$

Of the electrons which react with H_3O^+ a fraction, f_1' , oxidize I^- by reactions (3) and (4). Under these conditions the quantum yield of iodine is given by

$$\phi(I_3^-) = \phi_{e-(aq)} [f_N + f_1'(1 - f_N)] \quad (13)$$

Other relationships which follow from the stoichiometry are:

$$\begin{array}{l} \phi'(I_3^-) = \phi'(N_2) + \phi'(H_2) \\ \phi'(N_2) = f_N \cdot \phi_{e-(aq)} \\ \phi'(H_2) = f_1'(1 - f_N) \phi_{e-(aq)} \end{array}$$

As the pH of the solution is reduced, f_1' approaches a value of unity while f_N decreases toward zero. Thus $\phi'(I_3^-)$, which at natural pH is equal to $\phi_{e-(aq)}$, decreases as the pH is reduced, goes through a minimum in the region of pH 0.5–1.0 and then increases again at high acid concentration (3). Over the same pH range $\phi'(N_2)$ decreases toward zero while $\phi'(H_2)$ increases from zero to a value approaching $\phi_{e-(aq)}$. These effects are illustrated in Figure 1.

If the yields of N_2 and H_2 are measured in the N_2O -saturated solutions the relative rates of reaction (2) and (9) can be determined unambiguously. Such measurements are not easily carried out in a student laboratory, but the relative rates of these two reactions may be estimated by assuming that the fractions f_1 and f_1' are equal at any particular acid concentration. If the conversion of $e^-(aq)$ to H occurs sufficiently close to the site at which the quantum is initially absorbed, the back-reaction(s) (7) will mainly involve an H atom and an I atom or I_2^- ion which stem from the same primary act. If these back-reactions are assumed to involve only this type of geminate recombination the relative rates of (3) and (7) depend only on the acid concentration irrespective of the presence or absence of N_2O , i.e., $f_1' = f_1 = [\phi(I_3^-)/\phi_{e-(aq)}]$ at all pH values. Substituting for f_1' and f_N in eqn. (13) one obtains eqn. (14):

$$\frac{k_9}{k_2} = \frac{\phi'(I_3^-) - \phi(I_3^-)}{\phi_{e-(aq)} - \phi'(I_3^-)} \cdot \frac{[H_3O^+]}{[N_2O]} \quad (14)$$

The ratio k_9/k_2 may thus be calculated from the yields of iodine in deoxygenated and N_2O -saturated solutions at a particular pH , knowing also the yield from an N_2O saturated solution at natural pH (where $\phi'(I_3^-) = \phi_{e-(aq)}$). Since the calculation involves only ratios of iodine yields it is unnecessary to determine absolute quantum yields. Equation (14) may thus be re-written as eqn. (15) where Y denotes the total yield of product after a fixed exposure time.

$$\frac{k_9}{k_2} = \frac{Y'(I_3^-) - Y(I_3^-)}{Y_{e-(aq)} - Y'(I_3^-)} \cdot \frac{[H_3O^+]}{[N_2O]} \quad (15)$$

The Experiment

Solutions containing 0.05 M NaI and various con-

centrations of HClO_4 are exposed to the 2537 Å radiation from a mercury resonance lamp. Such lamps also emit some light at 1849 Å which may be removed by interposing a water-filled quartz cell in the optical path. Light of longer wave-lengths emitted by the lamp is not absorbed by the solutions.

Oxygen is removed from the solutions by bubbling oxygen-free nitrogen or nitrous oxide through them. The gas flow is maintained during the photolysis to keep the solutions stirred, thus preventing the accumulation of reaction products near the front face of the cell. Since the extinction coefficient of $\text{I}^-(\text{aq})$ at 2537 Å is $220 \text{ M}^{-1}\text{cm}^{-1}$, the actinic light is absorbed mainly in the first 2 mm of solution so that the gas bubbles do not affect the amount of light absorbed. The yield of iodine is determined from the change in optical density of the solution at the absorption maximum of I_3^- (3530 Å, $\epsilon = 26,400 \text{ M}^{-1}\text{cm}^{-1}$). Since the equilibrium constant for the formation of I_3^- ($K = [\text{I}_3^-]/[\text{I}_2][\text{I}^-]$) is 742 M^{-1} at 25°C, in 0.05 M NaI solution 97 per cent of the iodine is present as I_3^- .

To minimize the thermal oxidation of iodide in aerated acid solutions the samples are prepared by mixing equal volumes of 0.1 M NaI and an appropriate acid solution immediately before beginning the deoxygenation.

Apparatus and Materials

Reaction vessel: fused silica optical cell, 10 mm light path, 30 mm diameter, fitted with socket for gas bubbler (see Fig. 2a).

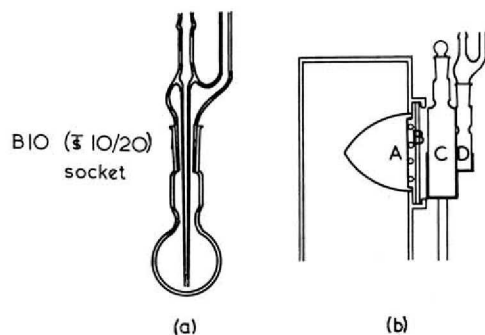


Figure 2. (a) Detail of reaction vessel and gas bubbler. (b) Photochemical arrangement: A, mercury resonance lamp; B, sliding shutter; C, water-filled silica cell; D, reaction vessel.

Water filter: fused silica optical cell, 20 mm light path, 5 to 7 cm diameter, filled with distilled water.

Low pressure mercury resonance lamp: preferably run from a stabilized power supply. A convenient source is a cold cathode discharge lamp in the form of a flat spiral approximately 8 cm in diameter, supplied by Engelhard Hanovia.

Spectrophotometer: for determination of iodine concentrations. The spectrophotometer cell-holder should be modified so that the reaction vessel may be inserted directly for the determination of optical densities.

Alternatively, a filter type photoelectric calorimeter may be used. A Chance OX-1 filter is suitable here but since the pass-band is so wide there are considerable deviations from Beer's Law and so it is necessary to calibrate the instrument using solutions of several known iodine concentrations.

Gas cylinders: cylinders of oxygen-free nitrogen (or argon) and nitrous oxide, each fitted with pressure reducer and needle valve and each having its own supply tubing for connection to the gas bubbler.

Solutions: 250 ml 0.1 M NaI stock solution. 100 ml each of solutions of HClO_4 , prepared by serial dilution and having nominal concentrations of 2.0, 0.6, 0.2, 0.06, 0.02, 0.006, and 0.002 M. One of these should be titrated with standard base to determine the actual concentrations. These solutions should be prepared from reagents of analytical quality. Ordinary distilled water may be used but it is preferable to use water which has been redistilled from alkaline permanganate.

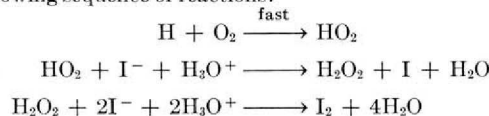
Procedure

The yield of iodine from an illuminated solution at natural pH and saturated with N_2O is first measured to establish the most suitable exposure time. Ten milliliters each of the NaI stock solution and water are mixed and the reaction vessel filled to the base of the stem with this solution. After passage of N_2O through the solution for ten minutes at a rate of 2–3 bubbles per second the gas bubbler is removed and the optical density of the solution is measured. The solution is then illuminated for short, timed intervals with N_2O bubbling through the solution during the illumination. The optical density of the solution is measured after each exposure. The exposure time for subsequent experiments is best taken as the time required to give an optical density of about 1.5 in this first experiment. The reason for choosing this rather high value is to minimize the effect of a residual trace of oxygen which causes a slightly faster rate of iodine production in the initial stages and especially at low pH² (see Fig. 3). In this first experiment, these reactions do not occur and the iodine yield is a linear function of exposure time. The yield after the chosen time of illumination may be most accurately obtained by linear extrapolation from shorter exposure times. In all subsequent runs at lower pH the yields are less and the concentration after the selected time of illumination can be measured directly with sufficient precision.

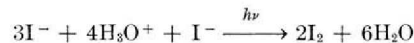
The sequence consisting of (1) saturation with N_2O , (2) measurement of optical density, (3) illumination for a fixed time, and (4) measurement of optical density, is carried out for a mixture of equal volumes of NaI stock solution and each of the acid solutions in turn.

The procedure is then repeated with a similar series

² The initial high yield of iodine in acid solution may arise from the following sequence of reactions:



Thus, if we include reactions (1) and (2), the net reaction is:



In an acid solution the initial iodine yield is therefore very sensitive to traces of oxygen.

An alternative procedure is to measure the concentration of I_3^- as a function of irradiation time for each solution as in Figure 3 and to use the slopes of the relatively linear final portions of the curves for the calculation of the relative rate constants. This would considerably increase the time required for the experiment.

of solutions using oxygen-free nitrogen in place of nitrous oxide.

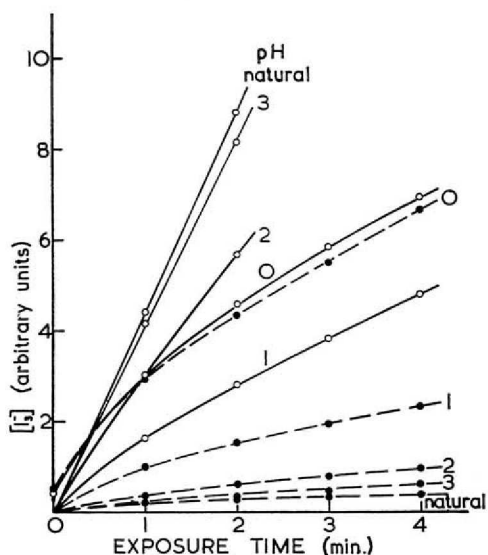


Figure 3. Iodine yields from the photo-oxidation of 0.05 M solutions of NaI at various pH values. Broken lines refer to solutions saturated with oxygen-free nitrogen. Full lines refer to solutions saturated with nitrous oxide. N.B. The finite $[I_3^-]$ at $t = 0$ for solutions of pH = 0 arises from thermal oxidation which occurs before deoxygenation.

Results and Discussion

The concentrations of I_3^- as a function of exposure time are shown in Figure 3. The yield after 2 min illumination is shown in Figure 4 and Table 1. The concentration of N_2O has been interpolated from the solubility data in Table 2, and used to calculate, as indicated below, the values of k_9/k_2 shown in Table 1.

For the solution with $[H_3O^+] = 1.0 \times 10^{-2} M$, $Y(I_3^-)$, and $Y'(I_3^-)$ are $(0.61 \pm 0.10) \times 10^{-5}$ and $(5.66 \pm 0.20) \times 10^{-5} M$ respectively and, from the first experiment, $Y_{e-(aq)}$ is $(8.80 \pm 0.30) \times 10^{-5} M$. Thus, from eqn. (15)

$$\begin{aligned} \frac{k_9}{k_2} &= \frac{5.66(\pm .15) - 0.61(\pm .10)}{8.80(\pm .30) - 5.66(\pm .15)} \times \frac{1.00 \times 10^{-2}}{2.70 \times 10^{-2}} \\ &= \frac{5.05 \pm .18}{3.14 \pm .35} \times \frac{1.00}{2.70} \\ &= 0.60 \pm .07 \end{aligned}$$

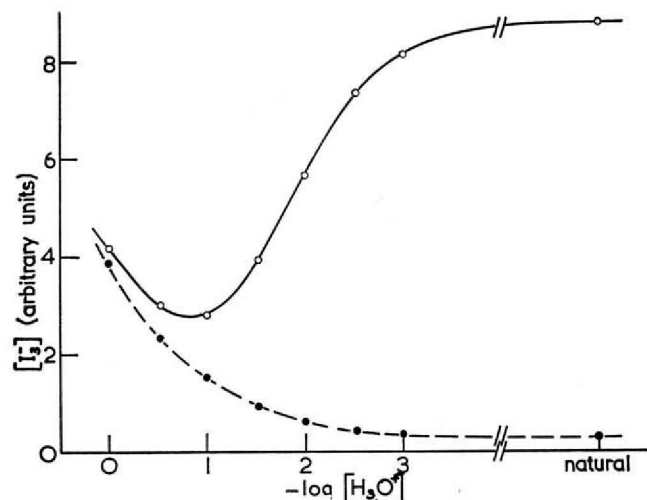


Figure 4. Iodine yields after 2 min illumination. Broken line: saturated with oxygen-free nitrogen. Full line: saturated with nitrous oxide.

(The errors in the solute concentrations should be less than 1% and are thus insignificant.)

Absolute values for the rate constants k_9 and k_2 have been determined by the techniques of pulse radiolysis (1). For the solutions used in this experiment the ratio k_9/k_2 was calculated to be about 0.6–0.7, increasing with acid concentration which increases the ionic strength. The increase to be expected in k_9/k_2 because of this effect is scarcely more than the experimental uncertainty, but the variation shown in the last column of Table 1 is approximately of the correct magnitude. For a further discussion of the effect of ionic strength see Ref (2) which also reports other definitive evidence of the nature of the reducing species. The good agreement between the results of this experiment and those from more rigorous investigations shows that the present method involves no undue over-simplifications.

Table 1. Iodine Yield for 2 min Illumination.

$[H_3O^+]$ (M)	$[I_3^-]$ (arbitrary units) after 2 min irradiation		k_9/k_2^a
natural	N_2O - saturated	Oxygen free	
(10^{-5})	8.80 ^b	0.28	...
1.0×10^{-3}	8.15	0.35	...
3.0×10^{-3}	7.35	0.43	0.53 ± 0.15
1.0×10^{-2}	5.66	0.61	0.60 ± 0.07
3.0×10^{-2}	3.93	0.93	0.68 ± 0.05
1.0×10^{-1}	2.80	1.52	0.79 ± 0.10
3.0×10^{-1}	3.01	2.34	...
1.0	4.17	3.88	...

^a Calculated from eqn. (15).

^b $Y_{e-(aq)}$

The experiment, as described here, has been in use for more than two years in our laboratories and has shown a high degree of reliability in student hands. There are very few photochemical experiments suitable for use in teaching laboratories and this experiment brings students into contact with very recent developments. The complexity of the reaction mechanism is

Table 2. Solubility of N_2O in Aqueous Solutions at a Gas Pressure of 760 mm (from the Data of Markham and Kobe (4) and of Watt (5)).

Temperature (°C)	$[N_2O]$ (M)
15	0.035
20	0.029
25	0.024
30	0.020
35	0.016

not necessarily a disadvantage in an experiment intended for advanced students; it may serve as a counter-balance to the illusion of simplicity which may be created by more elementary experiments.

Some modifications may be considered desirable in order to render the experiment more acceptable for a particular program. There are several which may be made with little change in the experimental setup and without encountering difficulties requiring deaeration and/or gas analysis.

If the time available for experimental work is short the study may be limited to solutions of natural pH in the presence and absence of N_2O . By varying the

initial concentration of I^- it can be demonstrated that the iodine yield is independent of $[I^-]$ providing that all of the photo-active light is absorbed in the solution.

Alternatively, the ratio of rate constants, k_9/k_2 , may be calculated from the results of fewer experiments, using only those acid concentrations at which a significant value can be determined. By so doing, the qualitative implications of the full curves in Figure 4 may be lost.

Absolute quantum yields may be calculated by the use of ferrioxalate actinometry; see Hatchard and Parker (6) for details of the experimental procedure. This requires a darkened room.

The effect of oxygen in the acid systems may be vividly demonstrated by bubbling oxygen through the solutions during illumination.

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